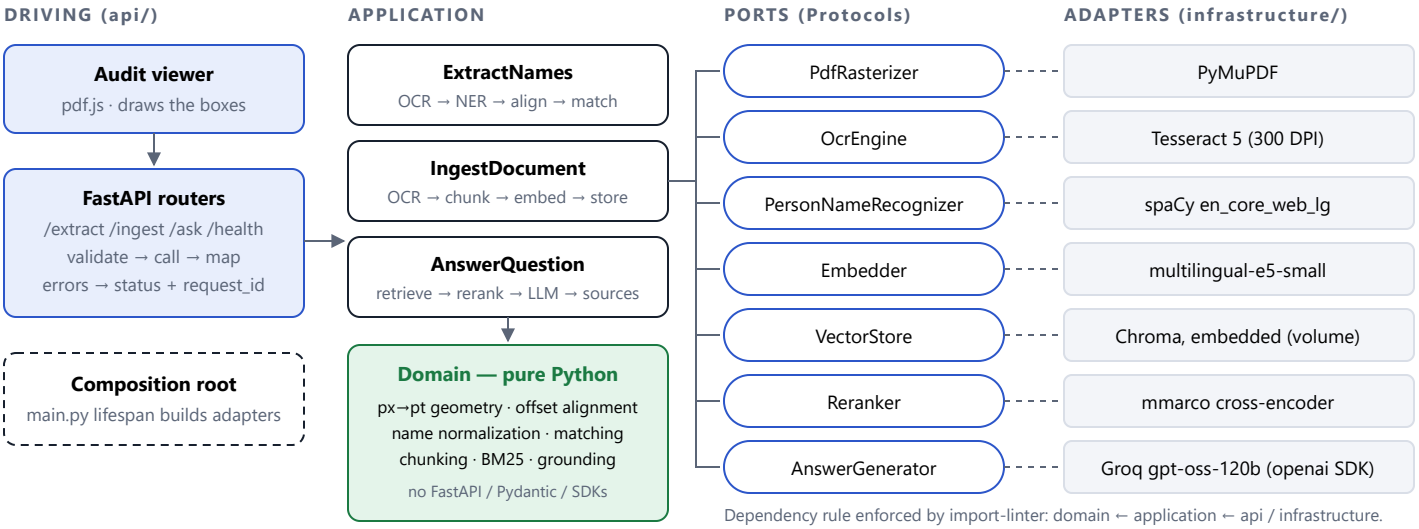
Document Intelligence Service — Design

Scanned PDF → every person name with its page and box in PDF points → fuzzy match against a requested list; plus RAG answers grounded in ingested documents. FastAPI · Python 3.12 · hexagonal architecture.

1.00 / 0.98	**54 / 54**	**11 / 12**	**15 / 15**	**0**
NER precision / recall on the OCR'd samples	boxes re-OCR'd from their crop = the name	retrieval hit@1 on golden questions	answers correct 3 runs in a row (pass^3)	sources not backed by a retrieved page

1 · Components

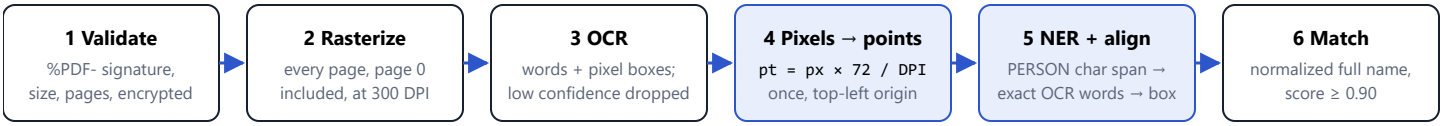
use cases depend on ports; each library lives in one adapter



Dependency rule enforced by import-linter: domain ← application ← api / infrastructure.

2 · Extract flow

POST /api/extract — what makes a box trustworthy



Real output (company_memo.pdf)

```
OCR words "Robert", "Chen" + their char offsets in the page text → spaCy PERSON span covers exactly those 2 words
→ union of their boxes → {page_number: 0, x: 59.76, y: 266.88, width: 53.04, height: 7.2} → matches "Robert Chen", score 1.0
```

Locating names by **character offsets** instead of searching the text gives one box per real occurrence, handles repeated names and never guesses. Matching is `max(ratio, token_sort_ratio)` on casefolded, accent-stripped names, so *María González* ~ *Maria Gonzalez* and word order is tolerated, with no substring inflation.

The README's sample stack (thefuzz) is replaced by rapidfuzz, which thefuzz now wraps. Pages are 0-based; coordinates rounded to 0.01 pt; a bottom-left consumer flips y' = page_height − y − height.

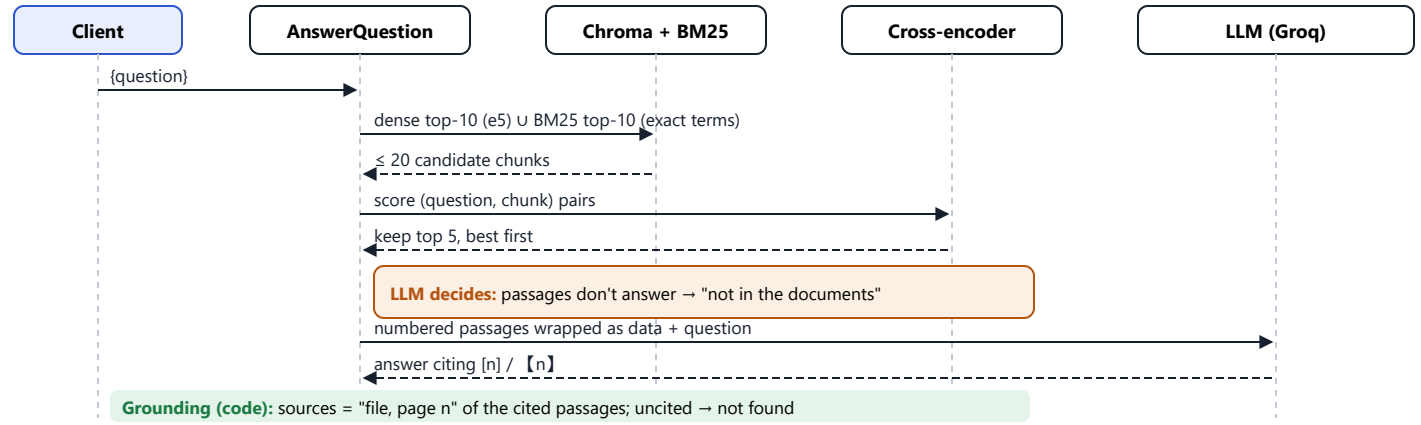
3 · Engineering practices

what the grader checks, where it lives

Configuration	One `Settings` object; every host, model, threshold, DPI and limit is a `DOCINTEL_*` variable (`.env.example`).
Validation	By content at the edge: `%PDF-`, empty, size, page count, encrypted/corrupt, malformed `names` — each a specific 4xx code.
Resilience	Domain errors → 4xx, infrastructure → 5xx with `request_id`; LLM timeout + retries; OCR per-page timeout.
Observability	Structured JSON logs with event names and `request_id`; names and document text never logged.
Tests	~870: pure domain, use cases on fakes, real-engine adapters, end-to-end on the samples; plus 5 eval scripts.

4 · Ask sequence

POST /api/ask — the LLM writes text; code decides everything else



Ingest. Same OCR → OCR layout blocks packed into ≤ 250-token, page-bounded chunks; a unit that does not fit moves *whole* to the next chunk, so no word is ever cut and nothing passes e5's 512-token window. IDs are `sha256(pdf)[:16]:chunk`: a second document can't overwrite the first and re-ingesting is an idempotent upsert.

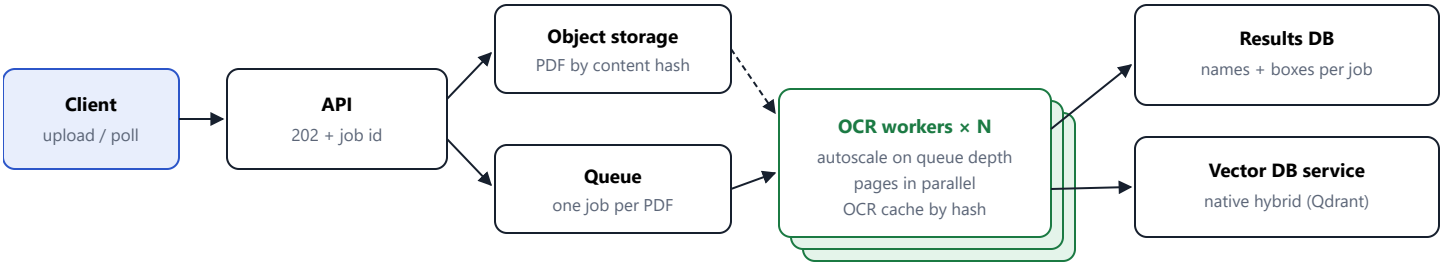
Why hybrid + rerank. Dense search misses exact names and figures; BM25 catches them. The reranker then puts the best five passages first. Whether they answer the question is the LLM's call: a cut-off on the reranker score was tried and dropped, because questions with and without an answer get overlapping scores.

5 · Technology choices and trade-offs

NEED	CHOICE	WHY	TRADE-OFF / REJECTED
OCR	Tesseract 5	Word boxes + confidence, light, CPU, eng+spa	Weak on poor scans → docTR / cloud OCR behind `OcrEngine`
PDF → image	PyMuPDF	Page size in points, rotation, CropBox	AGPL → pypdfium2 for commercial use
NER	spaCy `en_core_web_lg`	Only model with P/R ≥ 0.90 and zero ORG leaks; no torch	`sm` tagged "Atlas"/"Forbes" as people; `trf` needs torch
Matching	rapidfuzz	Full-string similarity, C++, MIT	thefuzz is a wrapper; partial_ratio inflates substrings
Embeddings	multilingual-e5-small	Spanish + English, 384-d, CPU	MiniLM is English-only; bge-m3 too heavy for CPU
Reranker	mmarco-mMiniLMv2-L12	Multilingual, 1.5 s / question on CPU	bge-reranker-v2-m3: 21 s on CPU (fine on a GPU)
Vector store	Chroma, embedded	Zero infrastructure (allowed by the brief); volume in Docker	Single writer; Qdrant/pgvector = one adapter away
LLM	Groq gpt-oss-120b	Any OpenAI-compatible endpoint by config; SDK timeouts + backoff	Free tier ≈ 30 req/min; data leaves the machine
Chunking, BM25	~30 lines each, in the domain	Pure and property-tested	LangChain / rank-bm25 add deps and hide graded logic

6 · Scaling to 1000+ PDFs/hour

1000 PDFs/h = 0.28 PDF/s. OCR dominates at ~2–3 CPU-s per page: a 5-page average keeps **~4 cores busy**, ~8 with headroom. The CPU work leaves the request path; the same image becomes the worker.



Also: GPU or hosted reranker/embedder · paid LLM tier with concurrency sized to its rate limit · OpenTelemetry traces per stage · lower DPI (200) as a speed/accuracy knob.

7 · Failure modes and mitigations

FAILURE	MITIGATION	RESPONSE
Not a PDF / empty / too big / too many pages / encrypted	Checked by content before any OCR	`4xx` specific `error.code`
Tesseract missing or stuck	Per-page timeout; error translated in the adapter	`503` `ocr_unavailable`
Poor scan, noisy words	Words below the confidence floor are dropped — no box over noise	`200` fewer names
LLM slow, rate-limited, down	SDK timeout + exponential backoff (honours Retry-After); extract/ingest unaffected	`503/502` `llm_*`
LLM invents a source	Impossible: sources come from cited passages that were sent; uncited → not found	`200` grounded
Question not covered	The LLM is told to say so; an answer without a citation counts as not found	`200` "not found"
Prompt injection in a document	Passages delimited as data; LLM has no tools; viewer renders plain text	`200` (golden probe passes)
Duplicate ingest / retry	Content-hash IDs → idempotent upsert	`200` no duplicates
ALL-CAPS documents (e.g. land registries)	Known limit: NER relies on capitalization → case-normalize before NER, or a stronger NER	documented

8 · Routing between models by query complexity

`AnswerGenerator` is a port, so a router is one more adapter — no use case changes. A **deterministic** router scores the question *after* retrieval:

- how many passages score high (1 → lookup, several → synthesis);
- cues for comparison, aggregation or "why";
- question length and language.

> **Simple lookup** → small fast model (gpt-oss-20b).
> **Multi-passage / reasoning** → large model (gpt-oss-120b).
> **Escalation:** if the small model returns no valid citation while retrieval scored strong evidence, retry once on the large model.

Thresholds are tuned on the golden Q&A set (pass^3 and cost per answer), never decided by the LLM itself.

Full record: TRD (docs/trds) with ADR-001…015 · decisions and AI-use disclosure: DECISIONS.md · run: `docker compose up --build` → http://localhost:8000/viewer